

# Peridoxia: A Python package for modeling redox evolution during mantle melting

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## Software

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## Summary

The redox conditions of the Earth's mantle have important implications for magma production across tectonic settings. Oxygen fugacity ( $fO_2$ ), a measure of the chemical activity of oxygen in a system, changes as a function of depth in the Earth due to the Earth's pressure and temperature gradient as well as changes in mineralogy. Within the upper mantle, the garnet-to-spinel transition exerts significant control over oxygen fugacity, as its depth lies near or within the pressure range of melting at common tectonic settings. While melts that form beneath mid-ocean ridges dominantly form in the spinel-field, hotspot settings likely experience significant garnet-field melting due to the higher potential temperatures imparted by the plume environment. Because the spinel and garnet mineral structures incorporate ferric iron differently, a peridotite melting in the garnet field may produce instantaneous melts with very different oxygen fugacity compared to a peridotite with the same bulk composition melting in the spinel field. The Peridoxia software package combines empirical partitioning data and thermodynamic activity-composition relationships to model the oxygen fugacity of mantle and melts during adiabatic decompression and associated melting.

## Statement of need

Peridoxia is an open-source Python package for modeling changes in oxygen fugacity during mantle decompression and melting, with applications across tectonic settings and through geologic time. Currently, the modeling of redox evolution during mantle melting is generally performed using thermodynamic modeling frameworks, such as pMELTS (Ghiorso et al., 2002), THERMOCALC (Powell et al., 1998), or Perple\_X (Connolly, 2009). These models use free energy minimization routines to predict stable mineral±melt assemblages and phase compositions. However, thermodynamic models are slow to update to new experimental results and often have shortcomings related to the handling of redox. For instance, pMELTS does not incorporate ferric iron ( $Fe^{3+}$ ) into the garnet phase, or chromium ( $Cr^{3+}$ ) into solid phases other than spinel, resulting in an overstabilization of spinel at high pressure. Additionally, most experiments that underpin the pMELTS model were performed in graphite capsules, making them less applicable to the higher oxygen fugacity conditions of the Earth's mantle. Experiments conducted at higher oxygen fugacity, that quantify ferric iron partitioning, are not incorporated into the pMELTS or PerpleX calibrations (e.g., Davis & Cottrell, 2018, 2021; Holycross & Cottrell, 2023; Sorbadere et al., 2018).

Peridoxia is designed to be an empirical (rather than thermodynamic) model that both incorporates recent experimental results and can be easily updated as new experimental data becomes available. Peridoxia is designed particularly to capture the nuances of partitioning

relationships between ferric iron in mantle phases. For instance, the temperature-dependence and compositional-dependence of ferric iron partitioning between spinel and liquid (Davis & Cottrell, 2021) are key parameters incorporated into the Peridoxia model.

## State of the field

Mantle oxygen fugacity has generated significant interest in recent years as analytical, experimental, and computational techniques for assessing redox-sensitive elements have improved. Current topics of particular interest to the community include the mechanism of oxidation at arcs (e.g., Cottrell et al., 2021; Evans & Tomkins, 2021; Muth & Wallace, 2022), reconciliation of proxies for oxygen fugacity at mid-ocean ridges (e.g., Birner et al., 2018; Davis & Cottrell, 2018; Nicklas et al., 2024), and the evolution of mantle oxygen fugacity through time (e.g., Aulbach & Stagno, 2016; Birner et al., 2024; Cottrell et al., 2025; Nicklas et al., 2019; Zhang et al., 2024).

Natural constraints on mantle oxygen fugacity are derived from volcanic rocks and exposed mantle lithologies. Volcanic rocks are abundant at Earth's surface but homogenize small-scale variations and do not evenly sample the mantle source. In contrast, exposed mantle lithologies are relatively rare and have invariably been altered by a combination of melting, melt-rock reaction, and fluid-rock reaction processes. Thus, computational models provide critical predictive power for interpreting the redox conditions of the upper mantle.

Computational thermodynamic models can predict  $fO_2$  given a bulk composition, temperature, and pressure, and recent studies have used these models to predict the redox evolution of a subsolidus peridotite assemblage. Jennings & Holland (2015) developed an improved thermodynamic model for assessing peridotite melting using the THERMOCALC framework and demonstrated that oxygen fugacity was highly dependent on phase assemblage—peridotites in the spinel field equilibrated at significantly lower oxygen fugacity for a given bulk rock redox ratio than peridotites in either the higher-pressure garnet stability field or lower-pressure plagioclase stability field.

Stolper et al. (2020) further investigated the low oxygen fugacity 'well' within the spinel stability field. Their modeling, performed in both pMELTS and PerpleX, demonstrated that the decrease was due to spinel's propensity for ferric iron. As pressure decreases from the garnet field to the spinel field, garnet breaks down to form spinel and pyroxene. Because garnet incorporates relatively high amounts of  $Al^{3+}$  and relatively low amounts of  $Fe^{3+}$ , the new spinel and pyroxene have low  $Fe^{3+}/Al^{3+}$  ratios. This results in a passive dilution effect that decreases  $fO_2$  as a peridotite ascends from the garnet field to the spinel field at constant bulk composition (constant bulk  $Fe^{3+}/Fe^{2+}$ ). A similar effect happens in reverse as a peridotite ascends further, from the spinel field to the plagioclase field. Because plagioclase incorporates negligible  $Fe^{3+}$  into its structure, the plagioclase-forming reactions concentrate ferric iron in the remaining ferric-bearing phases, resulting in an increase in oxygen fugacity despite zero change to the bulk redox state of the system.

Birner et al. (2024) developed the first version of the Peridoxia model (therein referred to as 'the empirical model') to assess redox evolution from the garnet field to the spinel field. The model was constructed using empirical partitioning relationships, in order to incorporate more nuanced partitioning data that are not captured by thermodynamic models. The results corroborated the decrease in  $fO_2$  from the garnet stability field to the spinel stability field and the corresponding ferric iron dilution mechanism. Birner et al. (2024) further demonstrated that the decrease in  $fO_2$  during the garnet-to-spinel transition has important implications for melting that begins in the garnet field. At high potential temperatures, such as occur in plume settings or early in Earth's history, melting may begin in the garnet field, where mantle oxygen fugacity is relatively high. The production of early high- $fO_2$ , high-temperature magmas efficiently extracts ferric iron, such that when the peridotite ascends into the spinel field, it retains little of its initial ferric iron and plummets to  $fO_2$  values significantly lower

92 than that of the spinel 'well' observed when melting does not occur. This mechanism may  
93 explain highly refractory, highly reduced peridotite residues observed at some mid-ocean ridges  
94 and has important implications for the oxygen fugacity of magmas produced during Earth's  
95 early history (Birner et al., 2024).

## 96 Software design

97 The goal of Peridoxia is to provide a user-friendly and intuitive framework for mantle melting  
98 calculations and analyzing model results. The Peridoxia API is organized around object-  
99 oriented Python modules and classes. The user interface is designed for flexibility and ease of  
100 use, with algorithmic details relegated to low level modules. The Peridoxia API is consistent  
101 across modules; for instance, accessing a property (such as temperature) of a single assemblage  
102 shares the same syntax as accessing that property across all assemblages within a model run.

103 The base modules of Peridoxia define the compositional parameters and partitioning  
104 constraints. Currently, the model contains seven oxide components (CaO, FeO, MgO, Al<sub>2</sub>O<sub>3</sub>,  
105 Cr<sub>2</sub>O<sub>3</sub>, Fe<sub>2</sub>O<sub>3</sub>, SiO<sub>2</sub>) and six phases that can form during a model run (olivine, orthopyroxene,  
106 clinopyroxene, garnet, spinel, and liquid). The compositions of each phase are constrained via  
107 partitioning relationships determined from natural and experimental data.

108 At each model step, material is transferred between phases to simulate processes such as melt  
109 production and the garnet-to-spinel phase transition. A set of exchange reactions then allows  
110 cations to redistribute between phases in order to maintain the required chemical partitioning  
111 relationships while maintaining mass balance.

112 A key innovation of the Peridoxia software package is the ability to calculate the activity of  
113 magnetite in spinel directly, using the activity-compositions of (Sack & Ghiorso, 1991b, 1991a)  
114 via integration of the ThermoEngineLite software package (Wolf et al., 2026). This allows for  
115 seamless calculation of  $fO_2$  within Peridoxia. Users can also select the Nell-Wood activity  
116 model (Nell & Wood (1991)) in place of the Sack-Ghiorso model.

117 The user-facing modules of Peridoxia allow the user to define the parameters of a model  
118 run, for instance, the potential temperature or the initial starting composition. After the  
119 completion of a model run, the results are stored to a ModelRun class object. This class  
120 contains properties and methods for probing the model results.

121 Peridoxia can be installed locally via pip (<https://pypi.org/project/peridoxia/>) or accessed in  
122 a light-weight virtual environment using MyBinder. The documentation ([https://peridoxia.  
123 readthedocs.io/en/latest/](https://peridoxia.readthedocs.io/en/latest/)) describes the model API and all model details and additionally  
124 contains a gallery with tutorials and examples that can be downloaded or run interactively in a  
125 web browser.

## 126 Research impact statement

127 The initial algorithms and parameters at the core of Peridoxia were developed for the  
128 manuscript 'Deep, hot, ancient melting recorded by ultralow oxygen fugacity in peridotites'  
129 (Birner et al., 2024). The model results were critical to the publication of that manuscript,  
130 providing a necessary comparison for thermodynamic model results. Critically, results from  
131 Peridoxia corroborated the thermodynamically-predicted structure of the  $fO_2$ -depth profile of  
132 the upper mantle and the ability for deep, hot, garnet-field melting to produce mantle residues  
133 and aggregated melts with low  $fO_2$  from a mantle with the same bulk oxidation state as today.  
134 While corroborating the broad results of the thermodynamic models, results from Peridoxia  
135 also highlighted issues with the handling of redox in these models—for instance, suggesting that  
136 the overstabilization of spinel in pMELTS may cause pMELTS to underestimate the magnitude  
137 of redox shifts between the garnet and spinel stability fields.

Since 2024, the model has been extensively refactored to create a faster, more robust, and more user-friendly software package. The new model also uses the ThermoEngine software package (Wolf et al., 2026) to automatically calculate the activity of magnetite in spinel. The updated version of the software was presented at the 2026 Goldschmidt Meeting (Birner et al., 2026) and has undergone extensive internal beta-testing.

## AI usage disclosure

Generative AI was not used at any point during the development of this software or the writing of this manuscript.

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